

Lab Exercise 11

Objective

The objective of this exercise is to enhance students' understanding of string handling, character classification, user-defined functions in the C programming language, and using standard string functions such as `strlen()`, `strcmp()`, `strcat()`, and character classification functions like `isdigit()` and `isalpha()`.

Task 1 : Username and Password Verification Using File

Write a C program that demonstrates the use of basic string handling functions.

The program should:

1. Declare three string: `str1[50]`, `str2[50]`, and `str3[50]`.
2. Read two strings from the user and store it in `str1` and `str2`.
3. Display the length of the first string using `strlen()`.
4. Copy the first string into another string using `strcpy()`.
5. Compare the two strings using `strcmp()` and display the comparison result.
If both string are equal, display "Both strings are equal" otherwise display "Both strings are NOT equal"
6. Concatenate the second string to the first string using `strcat()`.
7. Display clear output for each operation.

Sample output :

```
Enter first string - str1 : programming
Enter second string - str2: language

Length of first string: 11
After copy str1 to str3
Copied string (str3): programming
Both strings (str1 and str3) are equal.

After concatenation, str1 and str2 string: programminglanguage
The first string in UPPERCASE : PROGRAMMINGLANGUAGE
```

Task 2

Write a C program that verifies a **user's username and password** by comparing them with data stored in a file named **users.txt**.

The file stores records in the following format (one record per line):
username password

The program should:

1. Prompt the user to **enter a username and password**.
2. Read data from **users.txt** file.

The file contains a data as below

user.txt

admin 12345
user1 abc123
student pass789
turai 123\$ab

3. Compare the entered data with the file records using string comparison.
4. Display '**Login successful**' if a match is found.
5. Display '**Invalid username or password**' if no match is found.
6. Handle file opening errors properly.

Sample Input/Output

Enter username: user1
Enter password: abc123

Login successful

Task 3

Write a C program that performs **analysis and manipulation of a string** entered by the user. The program should accept a sentence and analyse its contents by identifying different types of characters. It should also demonstrate string manipulation by converting the sentence into uppercase letters.

Program Requirements

1. Declare a character array of size **100** to store a sentence.
2. Prompt the user to enter a sentence that may contain:
 - Alphabets
 - Digits
 - Spaces
3. Read the input **including spaces**.
4. Create **separate user-defined functions** for each of the following tasks:
 - Count the number of **vowels**

- Count the number of **consonants**
 - Count the number of **digits**
 - Count the number of **spaces**
 - Convert the string to **uppercase**
5. Display all results clearly with appropriate labels.
 6. Display the modified string after conversion to uppercase.

Functional Requirements

- Vowels include: a, e, i, o, u (both uppercase and lowercase).
- Consonants are alphabetic characters that are not vowels.
- Digits range from 0 to 9.
- Spaces refer to blank space characters (' ').
- The uppercase conversion must affect **all alphabetic characters** only.

Constraints and Guidelines

- Use the following header files where appropriate:
 - <stdio.h>
 - <string.h>
 - <ctype.h>
- Do **not** use global variables.
- Each function must return a value where applicable.
- The `main()` function should only handle input, output, and function calls.
- Follow proper indentation and meaningful variable naming.
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**Hint use built in function : `isalpha()`, `isdigit()`, `strchr()`, `isspace()`*

<https://www.geeksforgeeks.org/cpp/isalpha-isdigit-functions-c-example/>

Sample input/output

Enter a sentence: Hello World 123

String Analysis Results

```
Vowels      : 3
Consonants   : 7
Digits      : 3
Spaces      : 2
```

String in Uppercase:
HELLO WORLD 123